

Banks Response to Higher Capital Requirements: Evidence from a Quasi-Natural Experiment

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We study the impact of higher capital requirements on banks' balance sheets and their transmission to the real economy. The 2011 EBA capital exercise is an almost ideal quasi-natural experiment to identify this impact with a difference-in-differences matching estimator. We find that treated banks increase their capital ratios by reducing their risk-weighted assets, not by raising their levels of equity, consistent with debt overhang. Banks reduce lending to corporate and retail customers, resulting in lower asset, investment, and sales growth for firms obtaining a larger share of their bank credit from the treated banks. (*JEL* E51, E58, G21, G28)

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Basel III, which will become fully effective in 2019, significantly increases capital requirements for banks. However, at this point, the economic implications of such higher capital requirements are still unclear. Banks can, in

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principle, increase their regulatory capital ratios in two different ways: they can either increase their levels of regulatory capital (the numerator of the capital ratio) or they can shrink their risk-weighted assets (the denominator of the capital ratio) (Admati et al. 2018). While raising capital is generally considered “good deleveraging” by regulators, shrinking assets has potentially adverse effects if many banks simultaneously engage in cutting lending (Hanson, Kashyap, and Stein 2011). How banks adjust their balance sheets in response to higher capital requirements is thus an empirical question of crucial importance to understand the real implications of higher capital requirements.

The empirical identification of the effect of higher capital requirements on banks’ behavior faces a number of challenges. The most important challenge is to find exogenous variation in capital requirements. Yet, capital requirements tend to vary little over time, and when they do change, they change for all banks in a given economic area at the same time, leaving no cross-sectional variation to exploit. In the case when supervisors make use of discretion and impose bank-specific requirements, they will be correlated with (unobserved) bank characteristics and thus not be exogenous with regard to banks’ balance sheets. Finally, to assess the effects of capital requirements on bank lending, one needs to disentangle credit supply from credit demand.

We address these empirical challenges by exploiting the 2011 capital exercise, conducted by the European Banking Authority (EBA), as a quasi-natural experiment. The capital exercise required a subset of European banks to reach and maintain a 9% core tier 1 (CT1) capital ratio by the end of June 2012.¹ The institutional features of the capital exercise are particularly well-suited to address the above mentioned empirical challenges. First, the required CT1 ratio of 9% constituted an economically significant increase in capital requirements compared to the previously required 5%. Second, the rule by which banks were selected into the capital exercise allows us to disentangle the effect of capital requirements from effects associated with bank size. The EBA used a country-specific selection rule and included banks “in descending order of their market shares by total assets in each Member State” such that the exercise covered “50% of the national banking sectors in each EU Member State” (EBA 2011b). Since national banking sectors in Europe differ with regard to their total size, this country-specific selection threshold yielded a considerable overlap in size between banks selected and not selected into the exercise. Moreover, the explicit selection rule based on bank size implies that selection into the capital exercise was based on observable bank characteristics. We exploit this exogenous variation in the bank selection rule and employ a difference-in-differences matching estimation approach to examine how banks

¹ A bank’s CT1 capital ratio is defined as its CT1 capital over its risk-weighted assets, with CT1 capital comprising only the highest quality capital instruments (common equity), disclosed reserves, and hybrid instruments provided by governments (EBA 2011a).

subject to higher capital requirements adjust their balance sheets compared to otherwise similar banks not subject to a change in capital requirements.

Our main findings are as follows. First, we document that capital exercise banks (CEB)(our treatment group) raised their CT1 capital ratios by 1.9 percentage points more compared to banks not subject to the higher capital requirements (the control group). Capital exercise banks achieved this by reducing their levels of risk-weighted assets (RWA) by 16 percentage points. The control group is crucial for uncovering this finding: Capital exercise banks increased their levels of CT1 capital by 19% over our sample period, but the control group raised their levels of CT1 capital by the same magnitude.

We then investigate in detail how banks adjust both the asset- and liability side of their balance sheets in response to higher capital requirements. To study the effect on banks' balance sheet composition, we hand-collect information about banks' exposures to different asset classes from the banks' Pillar 3 disclosure reports. We find that treated banks mainly reduced their exposures to corporate and retail borrowers. In further tests, we show that capital exercise banks reduced their risk-weighted assets relative to the control group by engaging in asset shrinking rather than risk reduction. These results suggest that banks are reluctant to issue new equity to increase their capital ratios when required to do so by regulators. Potential explanations include asymmetric information and debt overhang. Admati et al. (2018) show that, in particular, banks with a large amount of outstanding subordinated debt should prefer asset sales to new equity issuances in the face of higher capital requirements. In line with this prediction, we show that capital exercise banks with an above median share of subordinated debt of total debt are more likely to shrink their assets and retire subordinated debt.

Simply observing a reduction in outstanding customer loans on banks' balance sheets is, however, not sufficient to conclude that the supply of credit by capital exercise banks contracted, since this might very well just reflect a reduction in credit demand by firms borrowing from capital exercise banks. To disentangle credit supply from credit demand, we use syndicated loan data and exploit the presence of multiple bank-firm relationships to control for credit demand. Specifically, we employ a modified version of the Khwaja and Mian (2008) estimator, which estimates the change in outstanding syndicated loans of a bank to country-industry firm clusters. We show that capital exercise banks reduced their credit supply of syndicated loans by 17 percentage points relative to banks in the control group.

Ultimately, the degree to which a reduction in credit supply from capital exercise banks implies real effects at the firm level depends on the extent to which other banks, not subject to higher capital requirements, "pick up the slack." We thus study, in a final step, whether the reduction in lending by capital exercise banks had real effects on firms. We find that firms with an initial high share of loans from capital exercise banks exhibited 4 percentage points lower asset growth, 6 percentage points lower investment growth, and

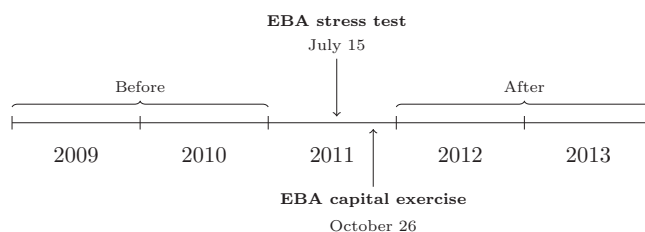
5 percentage points lower sales growth than firms less reliant on funding from capital exercise banks. This result is driven by unlisted firms which are less likely to substitute a reduction in credit supply with other sources of funding.

These results suggest that the 2011 EBA capital exercise had a detrimental impact on bank lending in Europe with adverse effects for the real economy, confirming concerns about a policy-induced credit crunch raised in advance by Acharya, Schoenmaker, and Steffen (2011), among others. Our findings therefore have important policy implications for bank capital regulation: if regulators (such as the EBA in the 2011 capital exercise) impose an increase in capital requirements on short notice and focus on capital ratios as the policy target variable, then banks will choose to deleverage via shrinking assets rather than via raising new equity. As suggested by Hanson, Kashyap, and Stein (2011), targeting the absolute amount of new capital that has to be raised instead of targeting the capital ratio could mitigate this problem, an approach which has been successfully applied in the U.S. stress test conducted in 2009 (Hirtle, Schuermann, and Stiroh 2009).

Our paper is most closely related to the literature examining the effect of shocks to banks' capital on bank lending. Peek and Rosengren (1997) exploit an exogenous shock to bank capital without a change in capital requirements to indirectly infer the effect on lending when capital requirements become binding. Another strand of literature seeks to directly exploit changes in capital requirements. An early study by Berger and Udell (1994) investigates bank lending before and after the introduction of Basel II, but without the benefit of exogenous cross-sectional variation in capital requirements.² To alleviate this concern, Kashyap, Stein, and Hanson (2010) adopt a model-based calibration approach for the United States, Fraisse, Lé, and Thesmar (2017) exploit variation in capital requirements across banks in France due to the use of internal risk models, Jimenez et al. (2017) analyze the introduction and later modifications in dynamic provisioning requirements in Spain, and Kisin and Manela (2016) estimate the shadow cost of capital requirements by exploiting a costly loophole that allowed banks in the United States to relax these constraints. Célérier, Kick, and Ongena (2016) explore the impact on lending in Germany by banks affected by tax reforms in Italy (in 2000) and Belgium (in 2006) which decreased their cost of bank equity. Finally, Mésonnier and Monks (2015) also exploit the EBA capital exercise and find that this regulatory event induced a credit crunch in the Euro Area.

We contribute to this literature in several ways. First, while most papers in the literature (with the exception of Mésonnier and Monks (2015)) study single-country settings, we exploit the country-specific bank selection rule of the 2011 EBA capital exercise to uniquely identify the effects of higher capital requirements across 18 countries. Second, our paper does not exclusively focus

² More recently, Aiyar et al. (2014), De Marco and Wieladek (2015), and Jensen (2015) exploit changes in bank-specific capital requirements in the United Kingdom and Denmark, respectively.

**Figure 1****2011 EBA capital exercise time line**

This figure shows the time line of the 2011 EBA capital exercise including the definition of the before and the after period used in the paper.

on lending, but investigates in detail how banks adjust both the asset and the liability sides of their balance sheets in response to an increase in capital requirements. Third, we examine why banks are reluctant to issue equity. We provide novel empirical evidence for the recent theoretical prediction by Admati et al. (2018) that banks' existing shareholders prefer to increase capital ratios by reducing assets rather than by raising new capital if banks can repurchase subordinated debt. Finally, we study the transmission of banks' balance sheet adjustments to the real economy in a multicountry setting and assess the resultant real effects on firms across Europe.

1. The 2011 EBA Capital Exercise

This section describes the objective and institutional details of the EBA capital exercise, which was announced by the EBA on October 26, 2011 (see Figure 1). The objective of the exercise was to restore confidence in the EU banking sector by ensuring that banks had sufficient capital to insure against unexpected losses. To achieve this objective, the EBA required 61 banks to build additional capital buffers to reach a 9% CT1 ratio by the end of June 2012. The capital exercise was an official "Recommendation" issued by the EBA. According to article 16(3) of the EBA regulation as established by the European Parliament, national supervisory authorities must make every effort to comply with the "recommendation." The EBA capital exercise did not coincide with other changes in capital requirements for European banks. In particular, the EU only started with the gradual introduction of Basel III in 2013 (Capital Requirements Directive IV). After the capital exercise, the EBA kept monitoring banks' compliance with the 9% CT1 ratio.

Both the timing and magnitude of this increase in capital requirements was unexpected. The capital exercise came only a few months after the EBA stress test in June 2011 and was described as a "quick-fire regulatory health check" (Halstrick and Framke 2011). The Financial Times reported that the 9% requirement was "well beyond the current expectations of banks and analysts" (Atkins, Jenkins, and Spiegel 2011). The credibility and rigor of the June stress

test had been criticized, in particular because the Belgian bank Dexia was declared in the stress test to be one of the safest banks in Europe, but had failed less than 3 months later (Greenlaw et al. 2012). Although both the EBA stress test and the subsequent EBA capital exercise increased capital requirements for capital exercise banks in 2011, the estimated 115 billion Euro capital shortfall due to the capital exercise dwarfed the 2.5 billion Euro capital shortfall due to the stress test (Acharya, Engle, and Pierret 2014). Thus, we naturally focus on the EBA capital exercise as the singularly overriding regulatory intervention.

The 61 capital exercise banks were selected based on total assets. In each country, the EBA included “banks in descending order of their market shares by total assets,” such that the exercise covered “at least 50% of the national banking sectors in each EU Member State in terms of total consolidated assets as of end of 2010” (EBA 2011b).³ Like in the 2011 EBA stress test, selection into the capital exercise was based on total assets as of end of 2010 and selection was therefore not based on bank-specific events in the months prior to the capital exercise.

Capital exercise banks were asked to submit their recapitalization plans to their respective national authorities outlining how they intended to reach the set targets. The EBA recommended that “banks should first use private sources of funding to strengthen their capital position to meet the required target, including retained earnings, reduced bonus payments, new issuances of common equity and suitably strong contingent capital, and other liability management measures.” The EBA also stressed that “reductions in risk-weighted assets due to the validation [...] should not, in general, be allowed as a means of addressing a capital shortfall unless these changes are already planned and under consideration by the competent authority” (EBA 2011c). However, the EBA also left discretion to the national supervisors which measures to take to enforce the higher capital requirements.⁴ In addition, the EBA did not specify how they would enforce their recommendations how to recapitalize.

2. Empirical Strategy and Data

This paper exploits the 2011 EBA capital exercise to identify how banks adjust their balance sheets in response to higher capital requirements and how this adjustment process affects firms which obtain a substantial share of their

³ From the initial 71 banks, the EBA excluded during the capital exercise banks which were “undergoing a deep restructuring,” namely Dexia, Österreichische Volksbank AG, West LB, all six Greek banks (EFG Eurobank Ergasias S.A., National Bank of Greece, Alpha Bank, Piraeus Bank Group, Agricultural Bank of Greece (ATE bank), TT Hellenic Postbank S.A., and Bankia. We do not include these banks in the analysis.

⁴ For example, “National supervisory authorities may, following consultation with the EBA, agree to the partial achievement of the target by the sales of selected assets that do not lead to a reduced flow of lending to the EU’s real economy but simply to a transfer of contracts or business units to a third party” (EBA 2011c). In contrast, the 2009 U.S. Supervisory Capital Assessment Program (SCAP) strictly required banks “to raise additional capital, either in public markets or by issuing mandatory convertible preferred securities” (Hirtle, Schuermann, and Stiroh 2009.)

borrowing from these banks. Hence, we first analyze at the bank level the extent to which the exercise changed bank behavior. Next, we move to the individual loan level to disentangle credit supply from credit demand. Finally, we examine the effect of higher bank capital requirements on asset, investment, and sales growth at the firm level.

2.1 Bank-level analysis

The setup of the capital exercise, whereby the EBA reviewed a subset of banks' actual capital positions and sovereign exposures and "requested them (i.e., our treatment group) to set aside additional capital buffers" (EBA 2011a), while leaving requirements unchanged for other European banks (i.e., our pool of control group banks), naturally lends itself to a difference-in-differences research design. However, selection into the capital exercise was not random. Instead, the EBA selected banks according to an explicit selection rule based on bank size, resulting in capital exercise banks being on average larger than noncapital exercise banks. This would compromise any causal inference if large banks would differ from small banks, for example in terms of business models or funding strategies, and would exhibit different trends even in the absence of a change in capital requirements.⁵ We exploit the country-specific selection threshold of the EBA selection rule in various ways to address this potential selection problem. Figure 2 shows the size distribution of capital exercise banks and noncapital exercise banks across different countries. While capital exercise banks are on average larger than noncapital exercise banks, the country-specific selection threshold yields a considerable size overlap between banks selected and not selected into the capital exercise. For example, while the smallest bank included in the EBA capital exercise, the Slovenian bank Nova Kreditna banka Maribor, had 6 billion euro in total assets as of end of 2010, the largest European bank not included in the capital exercise, the French bank *Crédit Mutuel*, had 591 billion euro in total assets in the same year. Knowledge about the selection rule based on observable characteristics (total assets) in combination with an overlap in size allows us to combine the difference-in-differences framework with an appropriate matching methodology by matching banks from the treatment group to similar banks from the pool of control group banks.

The paper uses the bias-corrected Abadie and Imbens (2011) matching estimator, which has recently been used by Almeida et al. (2011), Campello and Giambona (2013) and Kahle and Stulz (2013) in a corporate finance setting.⁶

⁵ The empirical setting of the capital exercise also seems to lend itself to a regression discontinuity design (RDD). Certain aspects of the empirical setting, however, preclude this approach from being the appropriate methodology. In the Online Appendix, we report the RDD estimation results in Table A5 and discuss the limitations of this approach due to the existence of multiple country-specific thresholds.

⁶ Figure 2 also shows that while the distributions of total assets overlap, they are significantly different. If the covariate distributions differ substantially, conventional regression methods can be sensitive to minor specification

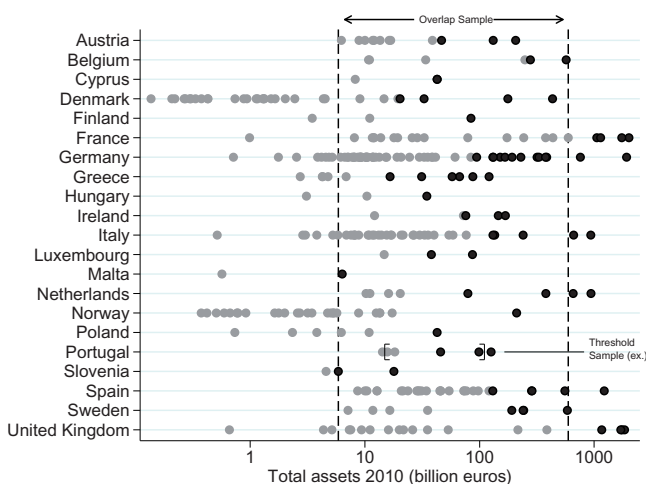


Figure 2
Bank size distribution by country

This figure shows banks size distribution (total assets as of end of 2010) of capital exercise banks (black) and noncapital exercise banks (gray) by country. The graph includes all ultimate parent banks headquartered in EBA supervised countries included in the SNL database. The figure also illustrates the construction of the overlap sample and the threshold sample. The overlap sample includes all banks larger than the smallest capital exercise bank (left vertical line) and smaller than the largest noncapital exercise bank (right vertical line). The threshold sample includes the two smallest capital exercise banks and the two largest noncapital exercise banks in each country (e.g., in Portugal).

To alleviate concerns that our results are driven by bank characteristics other than size, this paper also matches on pretreatment levels of the CT1 ratio, customer loans as a share of total assets, net interest income as a share of total operating revenue, depository funding as a share of total assets, and net income over total assets. These matching covariates capture potential differences in the capital structure, business models, funding strategies, and profitability of similarly sized banks prior to the capital exercise.

We adopt four different matching strategies, each exploiting the EBA's selection rule in a different way and each addressing a different identification concern. The *full sample matching* strategy matches four noncapital exercise banks to each capital exercise bank based on the six matching covariates using the full sample of 48 capital exercise banks and 144 noncapital exercise banks.⁷ Second, we match capital exercise banks to noncapital exercise banks in the “overlap sample” of banks which are larger than the smallest capital

changes because of their heavy reliance on extrapolation of regions where there is no support in the data (Imbens 2014). One approach to address this problem is the matching estimator developed by Abadie and Imbens (2011). Table A4 in the Online Appendix reports the results of the bank-level part of the paper using a regression-based approach.

⁷ Regarding the number of matches, we follow Abadie and Imbens (2011) and choose four matches, which was found to be a good trade-off between the bias (which is increasing in the number of matches) and the variance (decreasing in the number of matches) of the matching estimator.

Table 1
Matching strategies

Matching strategy	Full sample	Overlap	Within country	Within region
Sample used	Full sample	Overlap	Threshold	Threshold
Number of matches	1:4	1:1	1:2	1:4
<i>Matching covariates</i>				
Total assets	✓	✓	✓	✓
CT1 capital ratio	✓		✓	✓
Total deposits / Total assets	✓		✓	✓
Customer loans / Total assets	✓		✓	✓
Net int. inc. / Op. rev.	✓		✓	✓
Net income / Total Assets	✓		✓	✓
Country			✓	
Region				✓

This table displays the four matching strategies employed in the paper. The full sample includes 48 capital exercise banks and 144 noncapital exercise banks. The overlap sample includes all banks larger than the smallest capital exercise bank and smaller than the largest noncapital exercise bank. The threshold sample includes the two smallest capital exercise banks and the two largest noncapital exercise banks per country. The number of matches refers to the number of control group banks matched to each capital exercise bank. The matching covariate *Region* takes the value of 1 if the bank is headquartered in Greece, Ireland, Italy, Portugal, or Spain (GIIPS countries), and 0 otherwise.

exercise bank and smaller than the largest noncapital exercise bank. This *overlap matching* strategy completely removes the remaining size difference between capital exercise banks and noncapital exercise banks and rules out that our results are driven by bank size. Third, we match capital exercise banks to noncapital exercise banks around the selection threshold within the same country. Therefore, we construct a “threshold sample” that includes the two smallest capital exercise banks and the two largest noncapital exercise banks within each country. This *within-country matching* strategy addresses concerns that our results are driven by cross-country differences, such as regulatory interventions and different business cycles. Fourth and finally, we use the “threshold sample” and match capital exercise banks to noncapital exercise banks around the selection threshold within the same region (GIIPS countries and non-GIIPS countries). This *within-region matching* strategy specifically addresses the concern that our results are driven by the European sovereign debt crisis, which mainly affected banks in GIIPS countries (Acharya et al. 2016). For all four matching strategies, we estimate the average treatment effect on the treated (ATT) on banks’ outcomes using the bias-corrected Abadie and Imbens (2011) matching estimator. Table 1 provides an overview of our four matching strategies.

For the bank-level part of the paper, we use annual bank balance sheet data from the SNL Financial Company database. Our initial sample contains all 61 capital exercise banks and all 494 noncapital exercise European commercial and savings banks from the SNL Financial universe. Since the EBA capital exercise was conducted at the highest level of consolidation, we exclude all subsidiaries of capital exercise banks, noncapital exercise banks, and foreign banks. As the paper wants to track the behavior of independent banks over

Table 2
Summary statistics

Variable	Mean	Median	SD
<i>A. Bank-level analysis</i>			
Δ CT1 ratio	1.82	1.95	2.60
Δ log CT1 capital	0.19	0.19	0.23
Δ log tier 1 common equity	0.17	0.18	0.26
Δ log retained earnings	0.11	0.18	0.62
Δ log share capital & premium	0.23	0.10	0.48
Δ (Regulatory adjustments/CT1 capital)	-0.01	0.00	0.34
Δ log RWA	0.02	0.01	0.23
Δ log credit RWA	0.00	-0.01	0.23
Δ log corporate exposures	-0.16	-0.10	0.53
Δ log retail exposures	-0.04	-0.02	0.37
Δ log sovereign exposures	0.02	0.08	1.33
Δ log other exposures	0.38	0.35	1.01
Δ log market RWA	-0.16	0.03	1.18
Δ log operational RWA	0.12	0.09	0.28
Δ (RWA/total assets)	-4.34	-2.74	8.66
Δ (Loan loss reserves/customer loans)	1.42	0.41	2.76
Δ log total assets	0.11	0.09	0.21
Δ log customer loans	0.10	0.09	0.26
Δ log total securities	0.16	0.13	0.53
<i>B. Loan-level analysis</i>			
Δ log loan exposure	-0.02	0.00	0.68
<i>C. Firm-level analysis</i>			
Δ log total assets	0.10	0.07	0.34
Δ log fixed assets	0.11	0.05	0.43
Δ log sales	0.14	0.13	0.49

This table provides the summary statistics for all dependent variables used in the paper. Panel A provides the summary statistics for the dependent variables used in the bank-level analysis for the full sample (48 capital exercise banks and 144 noncapital exercise banks), panel B the summary statistics for the dependent variables used in the loan-level analysis and panel C the summary statistics of the dependent variables used in the firm-level analysis.

time, we also exclude all banks which were acquired during the sample period, all banks which received capital injections during the pretreatment period and all banks with negative levels of equity. This sample construction procedure finally leaves us with a sample of 48 capital exercise banks and 144 noncapital exercise banks.⁸ The sample period spans 2 post-treatment years after the capital exercise (2012 and 2013) and a symmetrical time window of 2 pretreatment years prior to the capital exercise (2009 and 2010).

To investigate how higher capital requirements affect the composition of banks' lending portfolios, we hand-collect the components of credit risk-weighted assets from the banks' Pillar 3 disclosure reports for the years 2009, 2010, 2012, and 2013 from the banks' Web sites and use these data to investigate for which exposure classes (corporate, retail, sovereign) banks adjust their credit risk-weighted assets. Panel A of Table 2 provides the summary statistics of all dependent variables used in the bank-level part for the full sample.

⁸ Table A1 in the Online Appendix lists all capital exercise banks in our sample.

2.2 Loan-level analysis

While bank balance sheet data are appropriate for investigating how banks adjust their balance sheets in response to higher capital requirements, it is not suitable for identifying the effect on bank lending. In particular, by using bank balance sheet data one cannot disentangle credit supply from credit demand. Thus, to study the effect of higher capital requirements on banks' credit supply, we use loan-level data on syndicated loans and, for identification, exploit multiple bank-firm relationships in the spirit of Khwaja and Mian (2008). As syndicated loans often have long maturities, bank exposures to individual firms are therefore often constant over time. We thus modify the estimator similar to Popov and Van Horen (2015) and Acharya et al. (2016) and aggregate firms into clusters based on their industry and country of incorporation. By clustering at the country-industry level, we ensure that firms are subject to the same regional and sectoral shocks over time and we attribute the remaining variation in loan exposure volumes to a reduction in credit supply.

We then estimate the following difference-in-differences regression specification:

$$\Delta \log \text{loan exposure}_{bij} = \beta \cdot \text{CEB}_{bi} + \gamma \cdot X_{bi} + \eta_i + \eta_j + \epsilon_{bij}, \quad (1)$$

where $\Delta \log \text{loan exposure}_{bij}$ is the change in loan exposures of bank b in country i to firm cluster j between the four quarters before the EBA capital exercise (2010Q3–2011Q2) and the four quarters after the capital exercise (2012Q3–2013Q2).⁹ The variable CEB_{bi} takes on the value of 1 if the bank is a capital exercise bank, and 0 otherwise. In addition, the specification includes bank characteristics as of 2010 (log total assets, CT1 ratio, customer loans as a share of total assets, net interest income as a share of total operating revenue, depository funding as a share of total assets, and net income over total assets); firm-cluster fixed effects η_j , which absorb all cluster-specific credit demand shocks; and fixed effects for banks' home countries, η_i , to absorb shocks which affect all banks in a given country. Like Khwaja and Mian (2008), we follow Bertrand, Duflo, and Mullainathan (2004) and collapse our data into a single pretreatment and a single post-treatment period before differencing to produce standard errors that are robust to concerns of autocorrelation. In addition, standard errors are clustered at the bank level.

For the loan-level part of the paper, we obtain data from Thomson Reuters LPC's DealScan database, which contains detailed information on syndicated loan contract terms, loan types, and maturities. We collect data on all outstanding term loans and credit lines from banks in our sample to nonfinancial

⁹ Our analysis investigates a time window of four quarters after the capital exercise (2012Q3–2013Q2) and thus focuses on the short-run adjustments of banks' credit supply in response to higher capital requirements. While a full-fledged analysis of the long-run effects of higher capital requirements is beyond the scope of this paper, we explore such potential long-run adjustments in Table A11 and Figure A4 in the Online Appendix.

corporate borrowers incorporated in EBA countries.¹⁰ Of the 76 banks in our matched control group, 63 were active in the syndicated loan market during our sample period and are feasible to serve as control group banks in the loan-level part of the paper. DealScan contains full information on the loan allocation between syndicate members for about 32% of all loans. For the remaining 68%, we follow De Haas and Van Horen (2012) and divide the loan facility equally among all members of a syndicate. Our initial sample contains 10,829 syndicated loans from 109 banks to 5,693 companies. The LPC DealScan database contains the issuance of new syndicated loans at the time of origination. To employ our modified version of the Khwaja and Mian (2008) estimator, we transform the data and calculate the outstanding exposure of bank b in country i to firm cluster j in quarter q using the maturity variable contained in the database. In our main analysis, we focus on the intensive margin sample which includes only country-industry firm clusters to which capital exercise banks lend both before and after the capital exercise. Thus, this sample excludes country-industry firm clusters that entirely stop borrowing after or do not borrow prior to the capital exercise. The intensive margin sample includes 45 capital exercise banks and 27 noncapital exercise banks.¹¹ Panel B of Table 2 provides the summary statistics of all dependent variables used in the loan-level part for the full sample.

2.3 Firm-level analysis

In the final empirical step, we link the capital exercise banks' balance sheet adjustments to real outcomes at the firm level. A reduction in credit supply of capital exercise banks would not necessarily yield effects at the firm level if other banks, not subject to an increase in capital requirements, would pick up the slack. An increase in capital requirements for the subset of capital exercise banks would then not affect the total supply of credit to the real economy and would not affect firms' corporate policies.

To measure a firm j 's dependence on credit supply from capital exercise banks prior to the capital exercise, we construct the variable *CEB borrowing share*:

$$\text{CEB borr. share}_j = \frac{\sum_{i \in [\text{CEB}]} \frac{1}{4} \sum_{q=2010Q3}^{2011Q2} \text{outst. loans}_{ijq}}{\sum_{i \in [\text{All banks}]} \frac{1}{4} \sum_{q=2010Q3}^{2011Q2} \text{outst. loans}_{ijq}}, \quad (2)$$

where the numerator is the average amount of outstanding loans of firm j obtained from capital exercise banks over the four quarters prior to the capital exercise (2010Q3–2011Q2) and the denominator is the average amount of total outstanding loans of firm j obtained from all banks over the same period. For

¹⁰ For term loans and credit lines, we follow the variable definition of Berg, Saunders, and Steffen (2016).

¹¹ In Table A14 in the Online Appendix, we provide additional results on the extensive margin sample of firms.

firms in our sample which were not borrowing in the syndicated loan market in the period before the capital exercise (but in the period after the capital exercise), we assign a *CEB borrowing share* of zero, since those firms were not dependent on credit from capital exercise banks prior to the capital exercise. In the bank- and loan-level part, we restrict our analysis to banks from EBA countries. Since European firms might also borrow from banks incorporated in non-European countries, we now also include those banks when computing the *CEB borrowing share*. We then divide our sample of firms into “CEB-dependent firms” with an above median dependence on credit supply from capital exercise banks as measured by the *CEB borrowing share* (our treatment group), and “non-CEB-dependent firms” with a below median dependence on credit supply from capital exercise banks (our control group pool). Since CEB-dependent firms might differ from non-CEB-dependent firms along a number of important characteristics, we employ a difference-in-differences matching methodology analog to the one used in the bank-level part. We match firms on country of incorporation, industry as defined by the 1-digit SIC code, whether the firm is publicly listed or not, and pretreatment levels of the logarithm of total assets, tangibility, cash flow over total assets, net worth, EBITDA over total assets, and leverage.¹²

Like in the bank-level part of the paper, we estimate the treatment effect on the treated (ATT) using the Abadie and Imbens (2011) bias-corrected matching estimator.¹³ The main outcome variables are the change in the logarithms of total assets, fixed assets (as a measure of investment, following Campello and Larrain 2016), and sales between the period before the capital exercise (2009 and 2010) and after the capital exercise (2012 and 2013). All variables are winsorized at the 5% level.¹⁴ As we expect results to be stronger for firms which are less likely to substitute a reduction in credit supply with other sources of funding (e.g., issuing equity), we also split our sample into listed and unlisted firms and report results separately.

For the firm-level part of the paper, we use information on firms’ balance sheets and profit and loss statements from Bureau van Dijk’s Amadeus Financials database. The database additionally contains information on a firm’s country of incorporation, its SIC industry code, and whether the firm is publicly listed. We have access to the sample of firms classified as “very large,” “large,” and “medium-sized” by Amadeus. Since the DealScan database and the Amadeus database share no common identifier, we hand-merge the two datasets and additionally require nonmissing values on all relevant variables, which leaves us with a sample of 1,958 firms. Panel C of Table 2 provides the

¹² Table A2 in the Online Appendix summarizes the definitions of all variables.

¹³ Table A15 in the Online Appendix additionally reports results of a difference-in-differences regression analysis.

¹⁴ We follow Acharya et al. (2016) for the level of winsorization. In unreported robustness tests, we find similar results when winsorizing the variables at the 1% level.

Table 3
Pretreatment characteristics of banks

	#	TA	CT1 ratio	Dep/ TA	Loans/ TA	NII/ OR	NI/ TA
<i>A. Unmatched sample</i>							
CEB	48	454.31	9.86	40.93	56.73	60.42	0.39
Non-CEB	144	24.43	11.41	55.54	66.62	67.69	0.41
Δ		429.87***	-1.55**	-14.61***	-9.89***	-7.27**	-0.02
<i>B. Full sample matching</i>							
CEB	48	454.31	9.86	40.93	56.73	60.42	0.39
Control	48	107.14	10.30	47.89	64.80	64.62	0.41
Δ		347.17***	-0.44	-6.95**	-8.07***	-4.19	-0.02
<i>C. Overlap matching</i>							
CEB	36	161.32	9.98	41.97	59.78	61.95	0.40
Control	36	156.10	10.95	53.80	57.06	71.89	0.38
Δ		5.22	-0.96	-11.83**	2.72	-9.94*	0.02
<i>D. Within-country matching</i>							
CEB	25	320.88	9.96	43.51	59.08	58.80	0.40
Control	25	80.92	10.80	43.21	61.72	71.22	0.42
Δ		239.96**	-0.84	0.31	-2.64	-12.42**	-0.02
<i>E. Within-region matching</i>							
CEB	26	310.18	10.01	44.85	59.77	58.99	0.45
Control	26	180.49	9.95	47.63	64.12	59.39	0.50
Δ		129.69	0.07	-2.77	-4.35	-0.40	-0.04

This table provides pretreatment mean comparisons for bank characteristics of capital exercise banks (CEB), noncapital exercise banks (non-CEB), and matched control group banks (control). Panel A compares the mean values of the 48 capital exercise banks and 144 noncapital exercise banks in the unmatched sample. Panels B to E compare capital exercise banks to the sample of matched control group banks using the *full sample matching*, *overlap matching*, *within-country matching*, and *within-region matching* strategies, respectively. “#,” “TA,” “CT1 ratio,” “Dep/TA,” “Loans/TA,” “II/OR,” and “NI/TA” denote the number of banks, total assets, the CT1 ratio, total deposits over total assets, customer loans over total assets, net interest income over operating revenue, and net income over total assets as of 2010, respectively. Table 1 lists the matching covariates used for each matching strategy. The paper tests for differences in means using Welch’s *t*-test. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

summary statistics of all dependent variables used in the firm level part for the full sample.

3. Results

3.1 Bank-level results

3.1.1 Summary statistics. We first provide summary statistics before and after matching for the different matching strategies. Table 3 shows the pretreatment mean values of the matching covariates for capital exercise banks, noncapital exercise banks, and matched control group banks as of end of 2010, the year immediately prior to the capital exercise. We use Welch’s *t*-test to test for differences in means between the groups.

Panel A of Table 3 compares the 48 capital exercise banks with 144 noncapital exercise banks in the unmatched sample. As expected, capital exercise banks significantly differ from noncapital exercise banks along a number of

important dimensions. Due to the capital exercise being carried out on the largest banks in each country, the mean capital exercise bank is more than 18 times larger than the mean noncapital exercise bank. The two groups of banks also significantly differ in terms of their business models, with the mean capital exercise bank being less engaged in customer lending and generating less of its revenue from interest income than the mean noncapital exercise bank. Moreover, the mean capital exercise bank has a lower CT1 ratio and is significantly less reliant on customer deposits (i.e., more reliant on wholesale funding) than the mean noncapital exercise bank. These large differences between capital exercise banks and noncapital exercise banks regarding important characteristics emphasize the necessity of employing a matching procedure.

Panel B of Table 3 shows the mean values of capital exercise banks and control group banks based on our *full sample matching* specification. This matching procedure significantly reduces the differences between capital exercise banks and noncapital exercise banks. While capital exercise banks are still larger than matched control group banks, this matching procedure reduces the difference from capital exercise banks being more than 18 times larger to capital exercise banks being roughly 4 times larger. To address concerns that our results might be driven by bank size, we employ the *overlap matching* strategy. Panel C of Table 3 shows that this matching strategy reduces the size difference to 5 billion euro, which is statistically insignificant. Panel D of Table 3 shows the post-matching summary statistics for the *within-country matching* strategy, which addresses concern that our results are driven by country-specific factors, and panel E of Table 3 shows the post-matching summary statistics for the *within-region matching* strategy, which specifically addresses the concern that our results are driven by the European sovereign debt crisis and banks from GIIPS countries.¹⁵

3.1.2 Adjustment of CT1 ratios. We first examine whether capital exercise banks did indeed increase their CT1 ratios in response to higher capital requirements, and whether they did so via increasing their levels of capital (adjustment via the numerator) or via reducing risk-weighted assets (adjustment via the denominator). The underlying assumption of a difference-in-differences estimator requires that capital exercise banks and matched control group banks would follow a similar trend in absence of the treatment (“parallel trend assumption”). Figure 3 shows the evolution of mean CT1 ratios relative to 2010 for capital exercise banks and the matched control groups for each of the four matching strategies. Figures 4 and 5 show the evolution of mean CT1 capital and mean risk-weighted assets relative to 2010, respectively. As can be seen in panel

¹⁵ Although the matching strategies reduce the differences between the two groups of banks, some differences remain significant. To address this problem, we use the Abadie and Imbens (2011) bias-corrected matching estimator; doing so introduces a bias-correction term to remove the bias in the coefficients stemming from imperfect matches on continuous covariates.

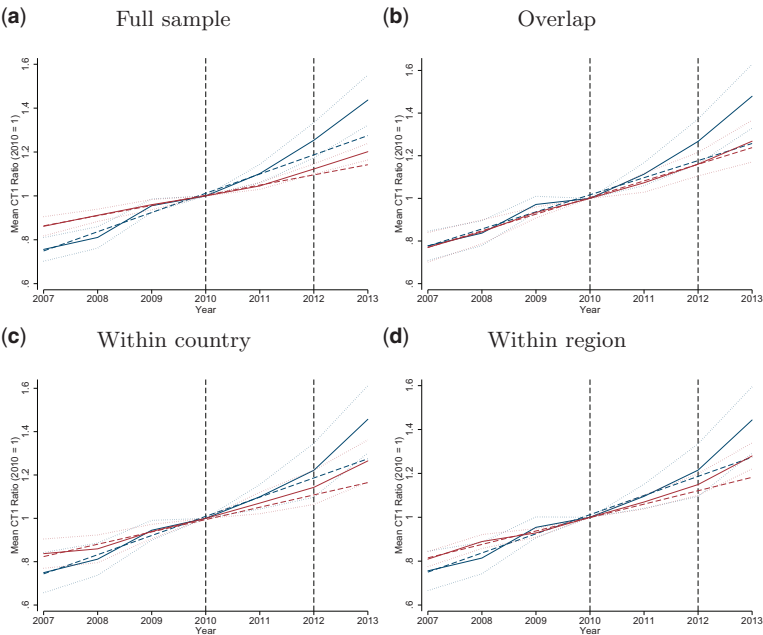


Figure 3
Core tier 1 ratios over time

This figure shows the evolution of the mean of core tier 1 (CT1) ratios over time for both capital exercise banks in the treatment group (solid blue line) and banks in the matched control group (solid red line) based on the four matching strategies. The two dashed vertical lines in each panel mark 2010 and 2012, the years immediately before and after the capital exercise. The dashed red and blue lines indicate the extrapolated pretreatment trends and the dotted lines indicated the 95% confidence intervals.

A of Figure 3, both capital exercise banks and matched control group banks in the full sample increase their CT1 ratios up to 2010, the year immediately prior to the capital exercise. Starting in 2011, capital exercise banks begin to increase their CT1 ratios significantly more than banks in the matched control group. Moreover, a comparison of the extrapolated pretreatment trends with the actual CT1 ratios shows that the capital exercise banks strongly diverge from their pretreatment trend after the capital exercise, while banks in the matched control group follow a fairly similar path in the post-treatment period. Panels B–D of Figure 3 show similar patterns for the other three matching strategies. As shown in Figure 4, capital exercise banks did not increase their CT1 ratios relative to the matched control groups by increasing their levels of core tier capital, but instead, as shown in Figure 5, by significantly reducing risk-weighted assets.

Table 4 reports formal tests for the differences in pretreatment trends between capital exercise banks and matched control group banks. As can be seen in panel A, capital exercise banks increased their CT1 ratios significantly more than matched control group banks in the full sample over the period from 2008 to 2010 due to a higher reduction in risk-weighted assets over this period. Panels

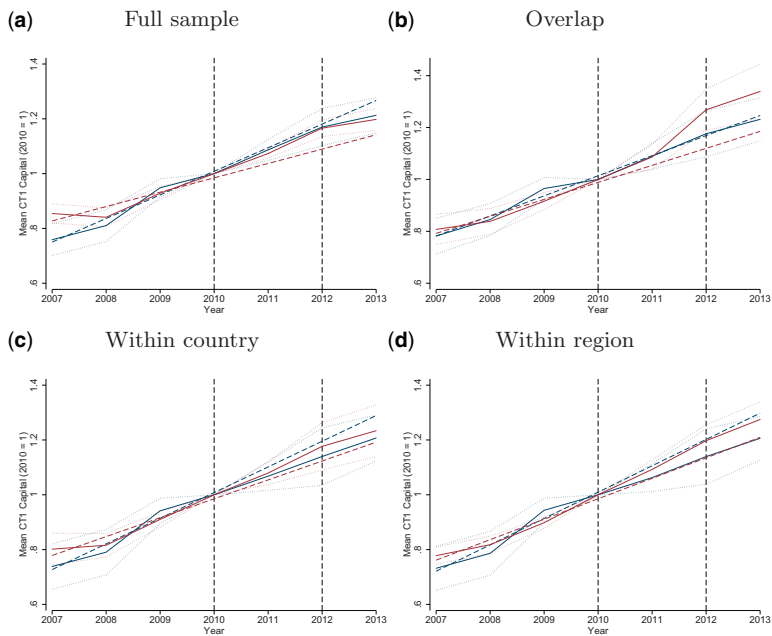


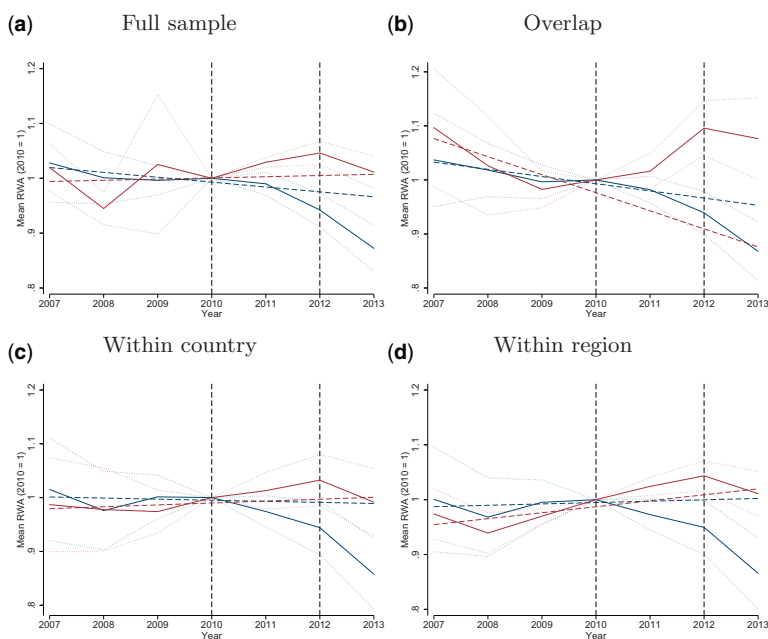
Figure 4
Core tier 1 capital over time

This figure shows the evolution of the mean of core tier 1 (CT1) capital over time for both capital exercise banks in the treatment group (solid blue line) and banks in the matched control group (solid red line) based on the four matching strategies. The two dashed vertical lines in each panel mark 2010 and 2012, the years immediately before and after the capital exercise. The dashed red and blue lines indicate the extrapolated pretreatment trends and the dotted lines indicated the 95% confidence intervals.

B–D of Table 4 show that the *overlap matching* and *within-country matching* strategies result in parallel pretreatment trends for CT1 ratios, CT1 capital, and risk-weighted assets, as can also be seen in panels B–D of Figure 4 and Figure 5. The advantage of these matching strategies is that they result in a comparison of more similar banks than in the full sample, at the cost of a smaller sample size. Thus, we report all results of the bank-level analysis for both the *full sample matching* strategy and the *overlap matching* strategy.¹⁶

Table 5 reports the estimation results for the changes in the CT1 ratios, in the logarithms of CT1 capital, and in the logarithms of risk-weighted assets from the period before to the period after the capital exercise between capital exercise banks and banks in the matched control groups. In each panel, row 1 reports the before-after differences for capital exercise banks, row 2 the before-after differences for matched control group banks, and row 3 the bias-corrected Abadie and Imbens (2011) matching estimator for the average treatment effect

¹⁶ For the sake of brevity, we report the results for the *within-country matching* strategy and the *within-region matching* strategy only for Section 3.1.2. All other results are available from the authors on request.

**Figure 5****Risk-weighted assets over time**

This figure shows the evolution of the mean of risk-weighted assets (RWA) over time for both capital exercise banks in the treatment group (solid blue line) and banks in the matched control group (solid red line) based on the four matching strategies. The two dashed vertical lines in each panel mark 2010 and 2012, the years immediately before and after the capital exercise. The dashed red and blue lines indicate the extrapolated pretreatment trends and the dotted lines indicated the 95% confidence intervals.

on the treated (ATT). The first column of panel A of Table 5 shows how both capital exercise banks and banks in the matched control group increased their CT1 ratios in the 2 years after the capital exercise, reflecting a general upward trend among European banks, which can also be seen in Figure 3. However, while matched control group banks increased their CT1 ratios by only 1.78 percentage points on average, capital exercise banks did so by 3.02 percentage points and thus significantly more than banks not subject to higher capital requirements. The ATT equals 1.86 percentage points and is significant at the 1% level, indicating that the increase in capital requirements did indeed affect the CT1 ratios of banks selected into the capital exercise.

The second column of panel A of Table 5 shows that capital exercise banks increased their levels of CT1 capital by 19% around the 2011 EBA capital exercise. However, as the comparison with the matched control group indicates, this increase seems to reflect a general development in the European banking system rather than an effect of the capital exercise. European banks not selected into the capital exercise exhibited an identical percentage increase in their levels of CT1 capital, rendering the ATT insignificant. This finding provides evidence

Table 4
Pretreatment trends in CT1 ratios

Variable	A. Full sample			B. Overlap		
	CEB	Control	Δ	CEB	Control	Δ
Δ CT1 ratio (10-09)	0.53	0.56	-0.03	0.40	0.70	-0.30
Δ CT1 ratio (10-08)	1.97	1.04	0.93***	1.76	1.73	0.02
Δ CT1 ratio (10-07)	2.41	1.71	0.70*	2.22	2.65	-0.43
Δ log CT1 cap. (10-09)	0.06	0.09	-0.03	0.04	0.09	-0.05*
Δ log CT1 cap. (10-08)	0.25	0.20	0.04	0.19	0.19	0.00
Δ log CT1 cap. (10-07)	0.31	0.20	0.11**	0.28	0.24	0.04
Δ log RWA (10-09)	0.01	0.03	-0.02	0.01	0.02	-0.01
Δ log RWA (10-08)	0.02	0.09	-0.07**	-0.01	0.00	-0.01
Δ log RWA (10-07)	0.00	0.02	-0.01	-0.01	-0.05	0.05
Variable	C. Within country			D. Within region		
	CEB	Control	Δ	CEB	Control	Δ
Δ CT1 ratio (10-09)	0.66	0.68	-0.01	0.56	0.75	-0.19
Δ CT1 ratio (10-08)	2.04	1.73	0.31	2.02	1.11	0.91*
Δ CT1 ratio (10-07)	2.66	1.97	0.69	2.60	1.85	0.74
Δ log CT1 cap. (10-09)	0.07	0.10	-0.03	0.06	0.12	-0.05**
Δ log CT1 cap. (10-08)	0.28	0.23	0.05	0.28	0.22	0.07
Δ log CT1 cap. (10-07)	0.34	0.27	0.07	0.35	0.28	0.07
Δ log RWA (10-09)	0.00	0.03	-0.03	0.01	0.04	-0.02
Δ log RWA (10-08)	0.05	0.04	0.01	0.05	0.08	-0.03
Δ log RWA (10-07)	0.01	0.05	-0.04	0.03	0.06	-0.03

This table presents the mean change in core tier 1 (CT1) ratios, the logarithms of CT1 capital, and the logarithms of risk-weighted assets for capital exercise banks and control group banks between 2010 and 2009, 2008, and 2007, respectively. The paper tests for differences in means using Welch's *t*-test. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

that capital exercise banks did not respond to the increase in capital requirements by raising new capital. In contrast, there is a significant difference in the change of risk-weighted assets between capital exercise banks and matched control group banks around the capital exercise, as can be seen in the third column of panel A of Table 5. While capital exercise banks reduced their levels of risk-weighted assets by 10 percentage points over the sample period, matched control group banks kept their levels of risk-weighted assets unchanged. The ATT indicates that capital exercise banks reduced their risk-weighted assets by 16 percentage points compared to banks in the matched control group which were not subject to an increase in capital requirements.¹⁷ The analog matching results of the *overlap matching* strategy in panel B, the *within-country matching* strategy in panel C, and the *within-region matching* strategy in panel D of Table 5 show that our results are robust to concerns of bank size, country-specific factors, and exposure to the European sovereign debt crisis, respectively.¹⁸ In particular, the results of the *within-region matching* strategy show that our

¹⁷ In the Online Appendix in Table A8, we furthermore show that these results are driven by weakly capitalized banks.

¹⁸ Note that in Table 5, the before-after differences do not always exactly add up to the ATT due to the bias-correction term introduced by the matching estimator. Figure A1 in the Online Appendix provides scatter plots of the dependent variables and shows that our results are not driven by a small number of outlier banks.

Table 5
Adjustment of CT1 ratios

Dependent variable	Δ CT1 ratio	Δ log CT1 capital	Δ log RWA
<i>A. Full sample matching</i>			
CEB: After - before	3.02***	0.19***	-0.10***
Control: After - before	1.78***	0.19***	0.03**
Bias-corrected ATT	1.86***	0.02	-0.16***
Number of observations	48	48	48
<i>B. Overlap matching</i>			
CEB: After - before	3.09***	0.18***	-0.11***
Control: After - before	2.40***	0.29***	0.08**
Bias-corrected ATT	0.72	-0.10	-0.19**
Number of observations	36	36	36
<i>C. Within-country matching</i>			
CEB: After - before	3.29***	0.17***	-0.12***
Control: After - before	2.34***	0.21***	0.01
Bias-corrected ATT	1.19**	-0.07	-0.15***
Number of observations	25	25	25
<i>D. Within-region matching</i>			
CEB: After - before	3.16***	0.17***	-0.11***
Control: After - before	2.15***	0.25***	0.04
Bias-corrected ATT	1.01**	-0.06	-0.13**
Number of observations	26	26	26

This table presents the estimates of the change in core tier 1 (CT1) ratios and its components. The dependent variables are the change in the CT1 ratio, the logarithm of CT1 capital, and the logarithm of the risk-weighted assets (RWA). In each panel, the first row contains the difference in the outcome variable for capital exercise banks (CEB) between the before (2009 and 2010) and the after (2012 and 2013) period, and the second row the difference in the outcome variable for matched control group (control) banks over the same period. The paper tests for differences-in-means using Welch's two-sample *t*-test. The third row contains the estimate for the average treatment effect on the treated (ATT) based on the bias-corrected Abadie and Imbens (2011) matching estimator. Panel A presents the results for the *full sample matching* strategy, panel B the results for the *overlap matching* strategy, panel C the results for the *within-country matching* strategy, and panel D the results for the *within-region matching* strategy. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

results are not driven by the European sovereign debt crisis which started in 2010 and mainly affected the economies of Greece, Ireland, Italy, Portugal, and Spain (GIIPS countries). This suggests that our findings also have external validity in situations in which banks are not under any particular financial distress (like in Northern Europe during our sample period).¹⁹ In all cases, the matching results suggest that capital exercise banks responded to the increase in capital requirements by reducing their risk-weighted assets compared to banks in the control group. The combined findings in Table 5 are the first central result of the bank-level analysis in our paper. They provide evidence that banks, when faced with an increase in capital requirements, adjust their capital ratios by reducing their levels of risk-weighted assets (adjustment via the denominator) rather than by raising new capital (adjustment via the numerator).

¹⁹ Table A6 in the Online Appendix provides a further regression-based test which shows that our results are not driven by banks from GIIPS countries. Additionally, Table A7 in the Online Appendix provides a placebo test around the start of the crisis in 2010 which shows that capital exercise banks and matched control group banks exhibited a similar evolution in their levels of CT1 capital and risk-weighted assets during this placebo period.

Table 6
Adjustment of CT1 capital components

Dependent variable	Δ log tier 1 Common Equity			Δ (Regulatory adjustments / CT1 capital)
		Δ log retained earnings	Δ log share capital & premium	
<i>A. Full sample matching</i>				
CEB: After - before	0.12***	0.04	0.16	−0.09**
Control: After - before	0.19***	0.14***	0.24***	0.01
Bias-corrected ATT	−0.08	−0.11	−0.06	−0.11**
Number of observations	48	45	43	48
<i>B. Overlap matching</i>				
CEB: After - before	0.13***	0.05	0.16	−0.06
Control: After - before	0.29***	0.14***	0.43***	0.00
Bias-corrected ATT	−0.16*	−0.09	−0.26	−0.06
Number of observations	36	34	34	36

This table presents the estimates of the change in the components of core tier 1 (CT1) capital. The dependent variables are the change in the logarithm of CT1 common equity, the logarithm of retained earnings, the logarithm of share capital and share premium, and the ratio of regulatory adjustments over CT1 capital. In each panel, the first row contains the difference in the outcome variable for capital exercise banks (CEB) between the before (2009 and 2010) and the after (2012 and 2013) period, and the second row the difference in the outcome variable for matched control group banks over the same period. The paper tests for differences-in-means using Welch's two-sample *t*-test. The third row contains the estimate for the average treatment effect on the treated (ATT) based on the bias-corrected Abadie and Imbens (2011) matching estimator. Panel A presents the results for the *full sample matching* strategy, and panel B the results for the *overlap matching* strategy. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

3.1.3 Adjustment of CT1 capital and risk-weighted assets components.

In this section, we study in detail the adjustments of the components of both CT1 capital and risk-weighted assets.²⁰ We supplement the SNL data on the components of CT1 capital and risk-weighted assets by hand-collecting missing data from the banks' annual reports. CT1 capital consists of tier 1 common equity (share capital and share premium plus retained earnings) and regulatory adjustments, which are deducted from tier 1 common equity. For example, goodwill and any other intangible assets are deducted from tier 1 common equity because of the high degree of uncertainty of their value in case of a default. Table 6 shows that both capital exercise and matched control group banks increased their tier 1 common equity by increasing their retained earnings and share capital, although matched control group banks did this at a faster rate. Instead, capital exercise banks reduced their regulatory adjustments more than the matched control group.²¹

²⁰ Table A3 in the Online Appendix shows a regulatory bank balance sheet and the decomposition of CT1 capital and risk-weighted assets used in this section.

²¹ The accounting rules governing these regulatory adjustments allow banks to manage these deductions to maximize their CT1 capital. Using a sample of U.S. banks, Lubberink (2014) shows that banks use these adjustments to increase their regulatory capital.

Table 7
Adjustment of risk-weighted assets components

	Credit risk-weighted assets			
Dependent variable	$\Delta \log \text{cRWA}$			
		$\Delta \log \text{corp. exposure}$	$\Delta \log \text{ret. exposure}$	$\Delta \log \text{sov. exposure}$
<i>A. Full sample matching</i>				
CEB: After - before	−0.13***	−0.23***	−0.10*	−0.04
Control: After - before	0.03***	−0.02	0.10***	−0.03
Bias-corrected ATT	−0.20***	−0.24**	−0.34***	0.88*
Number of observations	48	47	47	40
<i>B. Overlap matching</i>				
CEB: After - before	−0.13***	−0.24***	−0.08	−0.14
Control: After - before	0.06*	0.38***	0.19***	0.12
Bias-corrected ATT	−0.19**	−0.63**	−0.27	−0.24
Number of observations	36	35	35	28
	Market and operational risk-weighted assets			
Dependent variable	$\Delta \log \text{mRWA}$	$\Delta \log \text{oRWA}$		
<i>C. Full sample matching</i>				
CEB: After - before	−0.08	0.08*		
Control: After - before	−0.29***	0.09***		
Bias-corrected ATT	0.02	0.04		
Number of observations	46	47		
<i>D. Overlap matching</i>				
CEB: After - before	−0.24	0.07		
Control: After - before	0.30***	0.14***		
Bias-corrected ATT	−0.54	−0.07		
Number of observations	34	35		

This table presents the estimates of the change in the components of risk-weighted assets. The dependent variables are the change in the logarithm of credit risk-weighted assets (cRWA), consisting of credit risk-weighted assets for corporate exposures, retail exposures, and sovereign exposures, and the change in the logarithms of market risk-weighted assets (mRWA) and operational risk-weighted assets (oRWA). In each panel, the first row contains the difference in the outcome variable for capital exercise banks (CEB) between the before (2009 and 2010) and the after (2012 and 2013) period, and the second row the difference in the outcome variable for matched control group banks over the same period. The paper tests for differences-in-means using Welch's two-sample *t*-test. The third row contains the estimate for the average treatment effect on the treated (ATT) based on the bias-corrected Abadie and Imbens (2011) matching estimator. Panel A and C present the results for the *full sample matching* strategy, and panel B and D the results for the *overlap matching* strategy. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

A banks' risk-weighted assets consist of the risk-weighted assets for credit risk (cRWA), market risk (mRWA), and operational risk (oRWA). Table 7 presents the results for this decomposition of risk-weighted assets and shows that capital exercise banks reduced their risk-weighted assets for credit risk. This implies that capital exercise banks adjusted their loan portfolio, instead of their trading portfolio. Using hand-collected data from the banks' Pillar 3 disclosure reports, we further decompose the risk-weighted assets for credit risk into credit risk-weighted assets for corporate exposures, retail exposures (including exposures to SMEs), and sovereign exposures. Table 7 shows that the reduction in credit risk comes from a reduction in corporate and retail exposures.

Table 8
Risk reduction and asset shrinking

Dependent variable	Risk reduction		Asset shrinking		
	Δ (RWA/TA)	Δ (LLR/CL)	Δ log TA	Δ log CL	Δ log TS
<i>A. Full sample matching</i>					
CEB: After - before	-5.99***	1.56***	0.03	0.01	0.07
Control: After - before	-4.55***	1.29***	0.13***	0.10***	0.19***
Bias-corrected ATT	-0.71	0.64	-0.14***	-0.12***	-0.17**
Number of observations	48	48	48	48	48
<i>B. Overlap matching</i>					
CEB: After - before	-6.46***	1.94***	0.02	0.01	0.07
Control: After - before	-5.92***	1.97***	0.17***	0.28***	0.16*
Bias-corrected ATT	-0.64	0.00	-0.15*	-0.27**	-0.09
Number of observations	36	36	36	36	36

This table presents the estimates of the change in outcome variables associated with risk reduction and asset shrinking behavior. The dependent variables are the change in the ratio of risk-weighted assets over total assets (RWA/TA), the ratio of loan loss reserves over customer loans (LLR/CL), and the logarithms of total assets (TA), customer loans (CL), and total securities (TS). In each panel, the first row contains the difference in the outcome variable for capital exercise banks (CEB) between the before (2009 and 2010) and the after (2012 and 2013) period, and the second row the difference in the outcome variable for matched control group banks over the same period. The paper tests for differences-in-means using Welch's two-sample *t*-test. The third row contains the estimate for the average treatment effect on the treated (ATT) based on the bias-corrected Abadie and Imbens (2011) matching estimator. Panel A presents the results for the *full sample matching* strategy, and panel B the results for the *overlap matching* strategy. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

3.1.4 Risk reduction versus asset shrinking. Banks could reduce risk-weighted assets by changing the composition of their corporate and retail portfolios from riskier assets into safer assets, by recalibrating their internal risk-weight models (Behn, Haselmann, and Vig 2016), or by shrinking their assets. We construct two tests to examine which mechanism drives the reduction in risk-weighted assets. Both risk reduction and model recalibration would result in a lower average risk weight (risk-weighted assets/total assets) while keeping total assets constant. Pure asset shrinking would result in a constant average risk weight and a drop in total assets. Table 8 reports the matching estimation results for two different measures of banks' asset risk as the outcome variable. The first column shows that there is no statistically significant difference in the changes in the RWA/TA ratio between capital exercise banks and banks in the matched control group. Similarly, the second column shows that there is also no significant treatment effect with regard to loan loss reserves relative to outstanding customer loans. These results show that capital exercise banks did not reduce their risk-weighted assets by engaging in risk reduction.

Instead, Column 3 of Table 8 shows that capital exercise banks reduced total assets by 14 percentage points compared to banks in the matched control group. Moreover, the matching estimator in Column 4 of Table 8 indicates that capital exercise banks reduced outstanding customer loans by 12 percentage points compared to the matched control group of banks not subject to an increase in capital requirements. Finally, we also document a negative treatment effect on security holdings of capital exercise banks. However, as customer loans make

up 60% of the average capital exercise bank's balance sheet while security holdings only make up 27%, the asset shrinking behavior of capital exercise banks can mainly be attributed to a relative reduction in outstanding customer loans.

3.1.5 Why are banks reluctant to raise equity? We now turn to the question why banks are reluctant to raise equity. Potential explanations include debt overhang and asymmetric information. Admati et al. (2018) show that if banks can repurchase subordinated debt, existing shareholders find it preferable to deleverage by shrinking assets and repurchasing subordinated debt rather than by issuing new equity. The economic mechanism behind Admati et al. (2018) is a debt overhang problem: highly levered banks resist new equity issuances and may forgo positive NPV projects because the cash flows will accrue to debtholders. In a similar vein, Bahaj and Malherbe (2017) propose a theoretical model of bank behavior under capital requirements and also show that banks' lending response to an increase in capital requirements is more negative in the face of severe debt overhang.

A direct empirical implication of Admati et al. (2018) is that banks with higher levels of subordinated debt prefer asset shrinking and the repurchase of subordinated debt over a pure recapitalization.²² We test this prediction by splitting our sample into banks with above and below median levels of subordinated debt (9.3% hybrid securities and other subordinated debt of total debt) and separately study the effect of the capital exercise on the change in the CT1 ratio and its components in each subsample. Columns 1 to 3 of Table 9 show that capital exercise banks with above median levels of subordinated debt increased their CT1 ratio by reducing risk-weighted assets, while capital exercise banks with below median levels increased their capital ratios by increasing their levels of CT1 capital. This empirical finding is in line with the theoretical predictions of both Admati et al. (2018) and Bahaj and Malherbe (2017).

We furthermore test whether capital exercise banks with high levels of subordinated debt bought back their subordinated debt.²³ The evidence is somewhat mixed. Column 4 of Table 9 shows that while banks with high levels of subordinated debt reduce their holdings of subordinated debt by a large magnitude in the full sample, the coefficient is not significant, albeit large in magnitude, in the overlap sample.

Capital exercise banks could also be reluctant to issue new equity due to asymmetric information concerns. If investors interpret a bank's decision to issue equity as a signal that the bank's stock is overvalued, then banks might

²² See Proposition 9 (multiple classes of existing debt) in Admati et al. (2018).

²³ Vallée (2016) documents that numerous European banks bought back subordinated hybrid bonds trading under par value to strengthen their capitalizations.

Table 9
Banks with low and high holdings of subordinated debt

Dependent variable	Banks with low holdings of subordinated debt			
	Δ CT1 ratio	Δ log CT1 capital	Δ log RWA	Δ log Subord. debt
<i>A. Full sample matching</i>				
CEB: After - before	3.67***	0.25***	−0.08***	−0.24*
Control: After - before	2.11***	0.15***	−0.02	−0.06*
Bias-corrected ATT	2.75***	0.22***	0.00	0.10
Number of observations	26	26	26	26
<i>B. Overlap matching</i>				
CEB: After - before	3.99***	0.26***	−0.10***	−0.28*
Control: After - before	3.64***	0.17***	−0.06	−0.46***
Bias-corrected ATT	0.26	0.08	−0.03	0.18
Number of observations	20	20	20	20
<i>C. Full sample matching</i>				
CEB: After - before	2.25***	0.11***	−0.13***	−0.46**
Control: After - before	1.79***	0.23***	0.05**	−0.22***
Bias-corrected ATT	3.37**	−0.12	−0.59***	−0.51***
Number of observations	22	22	22	22
<i>D. Overlap matching</i>				
CEB: After - before	1.96**	0.09*	−0.13**	−0.53**
Control: After - before	2.29***	0.32***	0.09*	−0.26**
Bias-corrected ATT	−0.14	−0.22***	−0.23**	−0.31
Number of observations	16	16	16	16

This table presents the estimates of the change in core tier 1 (CT1) ratios, its components, and subordinated debt holdings for banks with low (panels A and B) and high (panels C and D) initial holdings of subordinated debt. Banks with low (high) holdings of subordinated debt are banks with a below (above) median (9.3%) share of subordinated debt of total debt as of 2010. The dependent variables are the change in the CT1 ratio, the logarithms of CT1 capital, risk-weighted assets and subordinated debt. In each panel, the first row contains the difference in the outcome variable for capital exercise banks (CEB) between the before (2009 and 2010) and the after (2012 and 2013) period, and the second row the difference in the outcome variable for matched control group banks over the same period. The paper tests for differences in means using Welch's two-sample *t*-test. The third row contains the estimate for the average treatment effect on the treated (ATT) based on the bias-corrected Abadie and Imbens (2011) matching estimator. Panels A and C present the results for the *full sample matching* strategy, and panels B and D the results for the *overlap matching* strategy. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

want to avoid sending out such a negative signal. Moreover, in the presence of debt overhang, the issuance of new equity might lead to a wealth transfer from existing stockholders to bondholders. We would therefore like to test how equity issuances by capital exercise banks and noncapital exercise banks affect the banks' stock and bond prices. We collect data on common equity issuances of banks in our sample from the SNL Capital Issuance Database and data on banks' stock and bond prices from Datastream. Yet, during the period of the capital exercise, only seven capital exercise banks and six control group banks announced equity issuances. Hence, it is difficult to draw strong conclusions from this analysis beyond the fact that seasoned equity issuances in the wake of the capital exercise were rare. However, the similar number of issuances between the two groups of banks provides additional evidence that

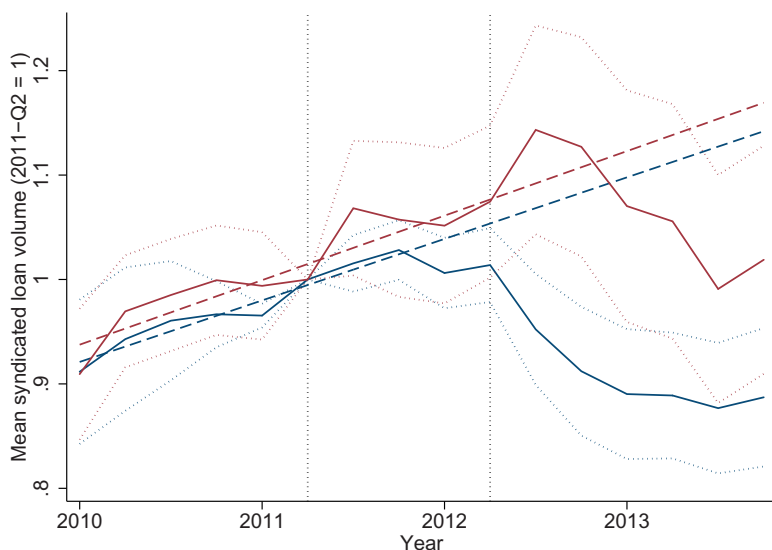


Figure 6
Syndicated lending over time

This figure shows the outstanding syndicated loan volume of capital exercise banks (solid blue line) and noncapital exercise banks (solid red line) over the period 2010Q1–2013Q4, normalized to one in 2011Q2. The two dashed vertical lines in each panel mark 2011Q2 and 2012Q2, the quarters immediately before and after the capital exercise. The dashed red and blue lines indicate the extrapolated pretreatment trends, and the dotted lines indicate the 95% confidence intervals.

any differential adjustment in CT1 ratios was unlikely to come from increases in the levels of equity.²⁴

3.2 Loan-level results

To rule out that the reduction in outstanding customer loans of capital exercise banks shown in Section 3.1 is driven by demand effects, we employ a modified version of the Khwaja and Mian (2008) estimator, which estimates the change in outstanding syndicated loan volumes of capital exercise banks and control group banks to Country $\times$ Industry firm clusters (see Acharya et al. 2016). Figure 6 shows the trends in outstanding syndicated loan volumes for capital exercise banks and control group banks relative to 2011-Q2, the quarter immediately prior to the capital exercise. There is a parallel upward trend in syndicated loan volumes of both groups of banks in the quarters leading up to the capital exercise. Starting in the third quarter of 2011, loan volumes of capital exercise banks started to stagnate and then decrease, while loan volumes for banks in the control group kept increasing.

²⁴ In Tables A9 and A10 in the Online Appendix, we present the results of short-term and long-term event studies for abnormal stock and bond returns and provide a more detailed discussion of this analysis. Our event study results also point toward debt overhang and not asymmetric information as the underlying economic reason capital exercise banks were reluctant to issue equity.

Table 10
Syndicated lending: Intensive margin

CEB	-0.09* (0.05)	-0.14* (0.08)	-0.16** (0.08)	-0.16** (0.07)	-0.17** (0.07)
Bank country FEs	Yes	Yes	Yes	Yes	Yes
Bank characteristics		Yes	Yes	Yes	Yes
Borrower country FEs			Yes	Yes	
Industry FEs				Yes	
Borrower country × Industry FEs					Yes
Capital exercise banks	45	45	45	45	45
Control group banks	27	27	27	27	27
Adjusted R^2	0.02	0.03	0.07	0.09	0.25
Observations	2,177	2,177	2,177	2,177	2,177

This table presents the estimation results of the change in lending around the 2011 EBA capital exercise from Equation (1) in Section 2.2:

$$\Delta \log \text{Loan Exposure}_{bij} = \beta \cdot \text{CEB}_{bi} + \gamma \cdot X_{bi} + \eta_i + \eta_j + \epsilon_{bij}$$

where $\Delta \log \text{Loan Exposure}_{bij}$ is the change in loan exposure of bank b in country i to firm cluster j between the four quarters before the EBA capital exercise (2010Q3–2011Q2) and the four quarters after the capital exercise (2012Q3–2013Q2). The variable CEB_{bi} takes on the value of 1 if the bank is a capital exercise bank, and 0 otherwise. Bank characteristics include log total assets, core tier 1 ratio, customer loans / total assets, net interest income / operating revenue, total deposits / total assets, and net income / total assets, all as of 2010. η_j are borrower country × industry (firm cluster) fixed effects, and η_i are bank country fixed effects. The intensive margin sample includes country-industry firm clusters to which banks lend before and after the capital exercise. Standard errors are clustered at the bank level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 10 presents the results of the difference-in-differences regression Equation (1) in Section 2.2 for the intensive margin sample. The first column of Table 10 shows that capital exercise banks reduced their exposures in the syndicated loan market by 9 percentage points after the capital exercise compared to banks in the control group. This specification includes fixed effects for banks’ home countries, which absorb unobserved shocks affecting all banks headquartered in a given country. The second column of Table 10 includes bank-specific control variables to address concerns that differences in bank characteristics are correlated with changes in credit demand, in particular bank size. In this specification, the magnitude of the coefficient increases to 14 percentage points. Credit demand shocks could conceivably also occur outside the bank’s home country. For example, Deutsche Bank might reduce its exposures to Spanish firms due to changes in credit demand in Spain.²⁵ Similarly, credit demand shocks could occur at the industry level and our results might be driven by capital exercise banks having different exposures to different industries than noncapital exercise banks. To address these concerns, we include borrower country fixed effects in the third column, industry fixed effects in the fourth column, and borrower country × industry fixed effects in

²⁵ In Table A13 in the Online Appendix, we further test whether banks reduced foreign lending significantly more than lending in their home country market, but we do not find evidence for such a “home bias” effect Giannetti and Laeven (2012).

the fifth column, respectively. In the fifth and strongest specification, which rules out that our results are driven by firm-cluster specific demand shocks, we find that capital exercise banks reduced their exposures in the syndicated loan market by 17 percentage points compared to banks in the control group.²⁶ This large negative effect of higher capital requirements on bank lending is in line with recent findings in the literature (Fraisse, Lé, and Thesmar 2017).

These results are consistent with the bank-level analysis in Section 3.1. Capital exercise banks responded to the increase in capital requirements by reducing outstanding corporate loans. The loan-level part of the paper shows that this reduction can be attributed to a reduction in credit supply and is not driven by demand effects.

3.3 Firm-level results

Ultimately, the degree to which a reduction in credit supply from capital exercise banks implies real effects at the firm level depends on the extent to which other banks, not subject to higher capital requirements, pick up the slack. To investigate whether such substitution occurs, we divide our sample of firms into “CEB-dependent firms” with an above median (65.0%) dependence on credit supply from capital exercise banks as measured by the *CEB borrowing share* (our treatment group), and “non-CEB-dependent firms” with a below median dependence on credit supply from capital exercise banks (our control group pool). Since CEB-dependent firms might differ from non-CEB-dependent firms along a number of important characteristics, we employ a difference-in-differences matching methodology analog to the one used in the bank-level part.

Table 11 shows the pretreatment mean values of the matching covariates for CEB-dependent firms, non-CEB-dependent firms, and matched control group firms as of end of 2010, the year immediately prior to the capital exercise. We use Welch’s *t*-test to test for differences in means between the groups.

Panel A of Table 11 compares the 952 CEB-dependent firms with 1,006 non-CEB-dependent firms in the unmatched sample. CEB-dependent firms are on average larger than non-CEB-dependent firms in terms of total assets, have a higher ratio of fixed assets to total assets (tangibility) and a higher leverage ratio. These differences between CEB-dependent firms and non-CEB-dependent firms emphasize the necessity of employing a matching procedure. We match four non-CEB-dependent firms to each CEB-dependent firm based on the Mahalanobis distance of all matching covariates as of end of 2010. This matching procedure renders all differences between CEB-dependent firms and non-CEB-dependent firms insignificant at the 1% level.

²⁶ Table A12 in the Online Appendix reports the results for credit line and term loan exposures separately. Although capital exercise banks also reduce their term loan exposures, our results are mainly driven by a reduction in credit line exposures. Since credit lines have shorter maturities than term loans, capital exercise banks seeking to reduce their risk-weighted assets could achieve this by not rolling over expiring credit lines.

Table 11
Pretreatment characteristics of firms

	#	log TA	Tang.	CF/ TA	Net worth	EBITDA/ TA	Lev.
<i>A. Unmatched sample</i>							
CEB firms	952	19.78	0.59	0.07	0.24	0.09	0.92
Non-CEB firms	1,006	19.48	0.56	0.07	0.25	0.09	0.91
Δ		0.30***	0.03***	0.00	-0.01	0.00	0.01**
<i>B. Matched sample</i>							
CEB firms	952	19.78	0.59	0.07	0.24	0.09	0.92
Control	952	19.80	0.59	0.07	0.24	0.10	0.92
Δ		-0.02	0.00	0.00	0.00	0.00	0.00

This table provides pretreatment mean comparisons for firm characteristics of CEB-dependent firms (CEB firms), non-CEB-dependent firms (non-CEB firms), and matched control group firms. CEB-dependent (non-CEB-dependent) firms are firms with an above (below) median (65.0%) share of their borrowing from capital exercise banks in the pretreatment period. “#,” “log TA,” “Tang.,” “CF/TA,” “Net worth,” “EBITDA/TA,” and “Lev.” denote the number of firms, the logarithm of total assets, tangibility, cash flow over total assets, net worth, EBITDA over total assets, and leverage as of 2010, respectively. Panel A compares the mean values of 952 CEB-dependent firms and 1,006 non-CEB-dependent firms in the unmatched sample. Panel B compares the 952 CEB-dependent firms to the sample of matched control group firms based on the bias-corrected Abadie and Imbens (2011) matching estimator. *, ** and *** indicated statistical significance at the 10%, 5%, and 1% levels, respectively.

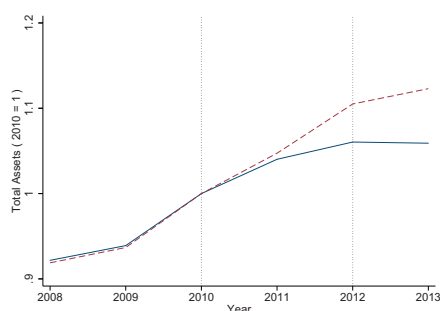
Figure 7 shows the evolution of total assets, fixed assets, and sales relative to 2010 for unlisted CEB-dependent firms and firms in the matched control group, respectively. Each of the panels shows that the corporate policies of CEB-dependent firms and non-CEB-dependent firms developed similarly up to 2010, the year prior to the capital exercise. Starting in 2011, CEB-dependent firms started to exhibit lower asset, investment, and sales growth than firms in the matched control group.

We estimate the differences in changes in the logarithms of total assets, fixed assets, and sales from the period before to the period after the capital exercise between CEB-dependent firms and firms in the matched control group. As we expect results to be stronger for firms which are less likely to substitute a reduction in credit supply with other sources of funding, we also split our sample into listed and unlisted firms and report results separately.

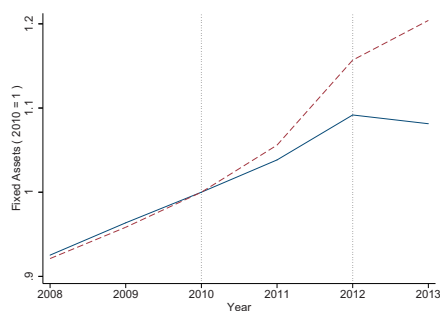
Panel A of Table 12 shows how the 2011 EBA capital exercise affected total assets, investment, and sales of all firms in our sample. Row 1 reports the before-after differences for CEB-dependent firms, Row 2 the before-after differences for matched control group firms, and Row 3 the bias-corrected Abadie and Imbens (2011) matching estimator for the average treatment effect on the treated (ATT). The average treatment effect shows that being dependent on funding from capital exercise banks had a significant negative effect on asset-, investment-, and sales growth. On average, CEB-dependent firms grew by 4 percentage points less, exhibited 6 percentage points less investment growth, and 5 percentage points less sales growth than firms in the matched control group less reliant on funding from capital exercise banks.

Panels B and C of Table 12 report results separately for the subsample of listed and unlisted firms, respectively. As expected, our results are driven by the

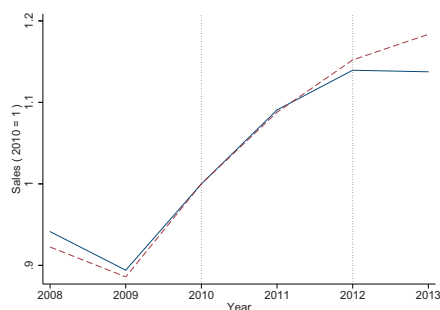
(a) Firm total assets over time



(b) Firm fixed assets over time



(c) Firm sales over time

**Figure 7****Firm-level outcomes over time**

This figure shows the evolution of the mean total assets (panel A), mean fixed assets (panel B), and mean sales (panel C) for both 681 unlisted CEB-dependent firms (solid blue line) and 793 unlisted non-CEB-dependent firms (dashed red line) firms in the matched control group, normalized to the value of 1 for the year 2010. The two dashed vertical lines mark 2010 and 2012, the years immediately before and after the capital exercise.

unlisted firms in our sample which are unable to raise public equity and thus have less alternative sources of funding. We find that unlisted CEB-dependent firms exhibited 6 percentage points less asset growth and 9 percentage points

Table 12
Firm-level outcomes

Dependent variable	$\Delta \log$ total assets	$\Delta \log$ fixed assets	$\Delta \log$ sales
<i>A. All firms</i>			
CEB firms: After - before	0.07***	0.07***	0.12***
Control: After - before	0.10***	0.10***	0.14***
Bias-corrected ATT	-0.04**	-0.06***	-0.05**
Number of observations	952	952	952
<i>B. Unlisted firms</i>			
CEB firms: After - before	0.04**	0.05*	0.14***
Control: After - before	0.10***	0.11***	0.06*
Bias-corrected ATT	-0.06***	-0.09***	-0.04
Number of observations	681	681	681
<i>C. Listed firms</i>			
CEB firms: After - before	0.13***	0.13***	0.04
Control: After - before	0.10***	0.08***	0.21***
Bias-corrected ATT	0.02	0.03	-0.06
Number of observations	271	271	271

This table presents the estimates of the change in firm-level outcomes around the 2011 EBA capital exercise. The dependent variables are the change in the logarithms of total assets, fixed assets, and sales. In each panel, the first row contains the difference in the outcome variable for capital exercise Bank (CEB)-dependent firms between the before (2009 and 2010) and the after (2012 and 2013) period, and the second row the difference in the outcome variable for matched control group (control) banks over the same period. The paper tests for differences in means using Welch's two-sample *t*-test. The third row contains the estimate for the average treatment effect on the treated (ATT) based on the bias-corrected Abadie and Imbens (2011) matching estimator. Panel A presents the results for all firms in our sample, panel B the results for unlisted firms in our sample, and panel C the results for listed firms in our sample. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

less investment growth than unlisted firms in the matched control group, while we find no significant difference for the sample of listed firms. Thus, our results show that the reduction in credit supply by capital exercise banks in response to higher capital requirements yielded significant negative effects for firms which obtained a large share of their funding from capital exercise banks. We conclude that the EBA capital exercise had negative effects on the real economy.

4. Conclusion

We exploit the EBA capital exercise as a quasi-natural experiment to study the effect of higher capital requirements on banks' balance sheet adjustments and the transmission of this effect to the real economy. Using different matching strategies which exploit the selection rule of the EBA capital exercise, we show that capital exercise banks increase their CT1 ratios more than noncapital exercise banks in response to an increase in capital requirements. This suggests that the capital exercise was an effective policy instrument to improve the capitalization of the largest European banks.

But the capital exercise may also have been a somewhat blunt instrument, because our analysis further shows that banks do not raise their capital ratios by increasing their levels of CT1 capital, but by reducing their risk-weighted

assets, in particular their credit exposures to corporate and retail clients. Consistent with debt overhang, we find that capital exercise banks with more subordinated debt are more likely to shrink assets and retire subordinated debt. As a consequence, we show that firms which are more reliant on credit supplied by capital exercise banks exhibit lower asset, investment, and sales growth than firms less reliant on capital exercise banks. This suggests that firms were unable to fully substitute the reduction in credit supply by capital exercise banks with other sources of financing.

An important policy implication of our paper is that capital requirements which target the regulatory capital ratio have potentially adverse effects on the real economy. As suggested by Hanson, Kashyap, and Stein (2011), targeting the absolute amount of new capital that has to be raised instead of targeting the capital ratio could mitigate this problem, an approach which has been successfully applied in the U.S. stress test conducted in 2009. In an institutional set up in which the recapitalization recommendations are difficult to verify and/or enforce, our paper highlights the risks associated with capital regulation that focuses on capital ratios as the policy target variable, while leaving it to the discretion of banks how to increase their capital ratios.

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